

Unit 2, Lesson 1: Introduction to Euclidean Geometry



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can distinguish between definition, postulate, and theorem.
- ☐ I understand the role of proof.
- ☐ I can relate properties of equality to the properties of congruence.



2.1.1 Warm-Up: Number Cubes Math

A picture of 16 number cubes is shown.



1. What do you notice?
2. Find the sum of the number cubes. Show your work or explain your thinking.
3. If all of the number cubes with 3 dots were changed to number cubes with 6 dots, how would you find the new sum? Show your work or explain your thinking.



2.1.2 Exploration: What is Euclidean Geometry?

Euclidean geometry is the study of planes and solid figures on the basis of axioms and theorems compiled by the Greek mathematician Euclid. Rather than using algorithms to solve problems like you may have done in other math courses, Euclidean geometry emphasizes using facts, ideas, and assumptions.



Euclid is known as the father of Geometry. He wrote 13 books using an axiomatic system to construct bodies of knowledge.

Each of Euclid's 13 books have proofs. Each proof shows how a hypothesis is true. The proof uses definitions, postulates, and proven theorems to explain why the hypothesis is true.

A **postulate** is an idea that is assumed to be true or something that is self-evident.

The first postulate in Euclid's Book 1 is shown.

A straight line segment can be drawn joining any two points.

1. Use a straightedge to show the postulate using the points shown.

R



• W

2. Discuss with your partner why a postulate is something that is self-evident. Summarize your discussion.

A **theorem** is an idea that has to be shown to be true through a chain of reasoning, often called a proof.

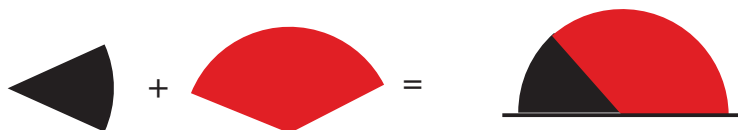
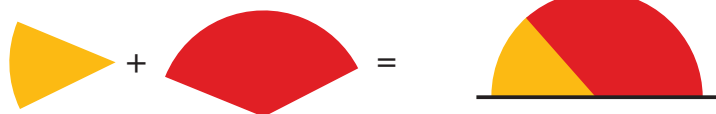


Vertical Angles Theorem: If two lines intersect, then vertical angles are congruent.

The vertical angles are



A pictorial proof of the vertical angles theorem is shown.



In the same manner,



3. Discuss with your partner the pictorial proof of the vertical angles theorem below. Complete the description for each picture shown in the pictorial proof.

Picture	Description
	The yellow angle and the red angle are being added.

Proofs play an important role in the course of geometry. Proofs can take on many forms. They are the set of logical steps that explain why something is true.

When programmers write a new game, they use a set of logical steps. When you tie your shoes, you use a set of logical steps.

4. Work with your partner to place the steps in tying a pair of shoes in order.

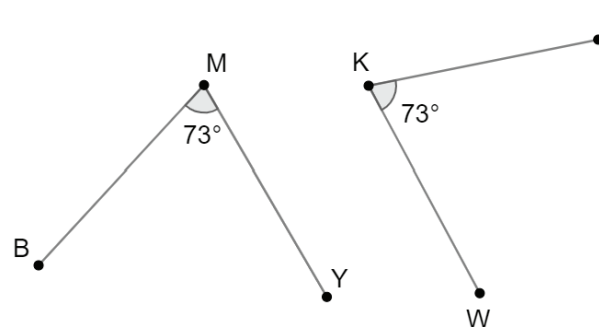
Step	Description	Picture
	Make a loop in one lace.	
	Cross the laces and pull them tight.	
	Pull both loops tight.	
	Wrap the other lace around the loop and push some lace through the hole that is made at the bottom to make a second loop.	

5. Discuss with your partner why someone who learned to tie their shoes using a different method may have difficulty putting these steps in a logical order. Summarize your discussion.



2.1.3 Guided Instruction: Properties of Equality and Congruence

In geometry, there are different notations for congruence and equality. For example, $\angle BMY$ and $\angle LKW$ have the same measure, so the angle measurements are equal. This is written as $m\angle BMY = m\angle LKW$. It can also be said that the angles are congruent, which is written as $\angle BMY \cong \angle LKW$.



Equality and congruence may seem to be the same, but equality is used when relationships can be expressed using numbers. Congruence is used when the relationship is not noted using numbers.

1. For each property, complete the example(s).

a. Reflexive property

Equality	Congruence
$m\angle WHD = m\angle \underline{\hspace{2cm}}$	$\angle WHD \cong \angle \underline{\hspace{2cm}}$

b. Symmetric property

Equality	Congruence
If $m\angle CKR = 59^\circ$ then $\underline{\hspace{2cm}}$.	If $\angle CKR \cong \angle HDE$, then $\underline{\hspace{2cm}}$.

c. Transitive property

Equality	Congruence
If $m\angle CRK = m\angle HWD$ and $m\angle HWD = 90^\circ$, then $m\angle CRK = \underline{\hspace{2cm}}$.	If $\angle CRK \cong \angle HWD$ and $\angle HWD \cong \angle MPX$, then $\angle CRK \cong \underline{\hspace{2cm}}$.

d. Substitution property

Equality
If $m\angle CRK + m\angle HWD = 156^\circ$ and $m\angle CRK = 84^\circ$, then $\underline{\hspace{2cm}} + m\angle HWD = 156^\circ$.

2. Write the conclusion for each statement. Then, name the property.

Statement	Property
If $m\angle FDT + m\angle BYH = 245^\circ$ and $245^\circ = m\angle MPW$, then	
If $m\angle FDT + m\angle BYH = m\angle RLC + m\angle KNW$ and $m\angle RLC = m\angle BYH$, then	

3. Discuss with your partner the conditional statement, "If $m\angle CRK = m\angle HWD$, then $\angle CRK \cong \angle HWD$." Summarize your discussion.

In geometry, there is a formal difference between equality and congruence. Equality refers to measurements, which is why $m\angle CRK = m\angle HWD$ includes the m for measurement. On the other hand, congruence means the shape is the same, which can be shown using isometry. In other words, the figure coincides exactly when superimposed.



If two figures have equal measurements, then by the **definition of congruence** the figures are congruent. If congruence is shown through a series of transformations, then the **definition of congruence in terms of rigid motions** was applied.

The definition of congruence can be used to change from congruence to equality.

Statement	Reason
1. $\angle CRK \cong \angle HWD$	1. Given
2. $m\angle CRK = m\angle HWD$	2. Definition of congruence

Or, the definition of congruence can be used to change from equality to congruence.

Statement	Reason
1. $m\angle CRK = m\angle HWD$	1. Given
2. $\angle CRK \cong \angle HWD$	2. Definition of congruence



Gottfried Wilhelm Leibniz (1646-1716) introduced a symbol using one horizontal bar under the tilde for congruence in an unpublished manuscript of 1679. In 1824, Carl Brandan Mollweide (1774-1825) used the modern congruent symbol with two horizontal bars under a tilde in the book, *Euklid's Elemente*.



2.1.4 Wrap-Up: Cool Down

1. Complete the table.

The terms and information used in the Exploration and Guided Instruction that I knew were...	Something I wondered during the Exploration or Guided Instruction was...	Something I learned during the Exploration or Guided Instruction was...